Uncertainty and Disturbance Estimator-Based Robust Control of Quadrotor without Velocity Measurements

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***Abstract*—In this paper, an uncertainty and disturbance estimator (UDE)-based position controller for quadrotors in the absence of velocity measurements is presented. Firstly, the nonlinear dynamics of a quadrotor is derived and simplified into a decoupled control design model using small angle approximation. Secondly, an UDE is constructed to compensate the system uncertainties and external disturbances. By applying passivity techniques to produce damping in the system, the proposed UDE-based controller can achieve trajectory tracking without velocity measurements and has a good robust performance. Moreover, the stability analysis of the closed-loop system is also introduced. Finally, the proposed method and the control method based on classical uncertainty and disturbance estimator are implemented and compared in the Gazebo simulation and flight experiment with a real quadrotor platform, are implemented and compared in the Gazebo simulation and flight experiment with a real quadrotor platform. Both simulation and experiment results demonstrate the effectiveness and advantages of the proposed control method in signal tracking and disturbance rejection.**

Keywords—disturbance rejection, quadrotor control, passivity filters, flight experiment

I. INTRODUCTION

Over the last years, unmanned aerial vehicles (UAVs) have received considerable attention from both academic communities and practical engineering for their potential prospects in various applications [1]-[4]. As one of the representation among UAVs, quadrotor is widely used in many fields due to their advantages, such as low cost, small size and easy to repair. This fact drives research institutes to the study of new control strategies, as any improvement in the quadrotor controller may deliver a more stable and robust flight performance.

Due to the importance of the robust control design in a quadrotor system, a few of research communities have studied the problem. In the past few years, numerous robust control methodologies have been applied to quadrotors [5], active disturbance rejection control (ADRC) [6], [7], adaptive control [8], [9], uncertainty and disturbance estimator (UDE) [10]-[12], and disturbance-observer-based control (DOBC) [13]. Compared to the robust control methods mentioned above, the UDE-based controller first proposed in [14], is relatively simple and computationally efficient. The basic idea of the UDE method allows approximating the engineering signal (the model uncertainty and external disturbance) by putting it through a filter with system states feedback in the frequency domain. However, in practice, the control systems may encounter the scenarios in which the velocity measurements of the quadrotor are either unavailable or not directly usable due to communication delays, measurement noises, etc. This limits the quadrotor application of the classic UDE-based controllers where velocity measurements are always required. Inspired by [15], [16], we aim to apply the passivity techniques in UDE-based robust control approach to quadrotor systems. A passive filter is constructed to deal with the unavailabilities of the velocity measurements.

In this paper, we develop an uncertainty and disturbance estimator (UDE)-based position controller for quadrotors in the absence of velocity measurements. The main contributions of this paper are summarized as follows:

1) An uncertainty and disturbance estimator is constructed to handle the system uncertainties and external disturbances.

2) The passivity techniques is applied in the controller design to avoid using velocity measurements

3) The proposed controller is validated in the Gazebo simulation and Flight Experiment using PX4 firmawre.

The rest of this paper is organized as follows. Section II presents the quadrotor dynamic model with external disturbance. Section III presents the proposed UDE-based control law, and the stability analysis is given in Section IV. The simulation and experiment results are given in Section V. Finally, Section VI concludes this paper.

II. QUADROTOR SYSTEM MODELLING

Consider a quadrotor vehicle represented by a rigid body of mass m, as shown in Fig. 1.

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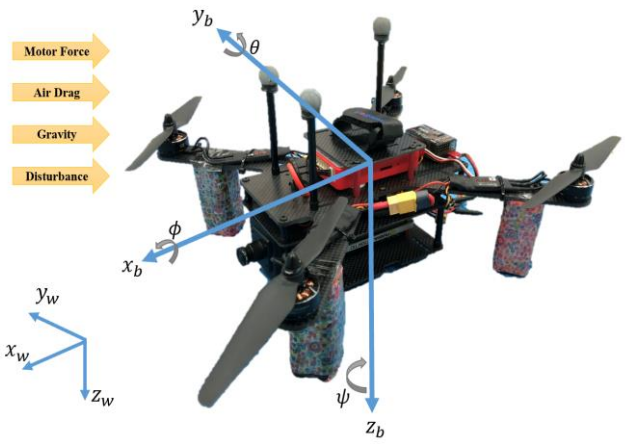


Fig. 1. The quadrotor with corresponding frames and forces.

e $\sum_w$ is the world inertial reference and $\sum_b$ is the body-fixed frame attached to the quadrotor at the mass center. $\boldsymbol{p}=[x,y,z]^T$ and $\boldsymbol{v}=[v_x,v_y,v_z]^T$ are, respectively, the position and linear velocity of the quadrotor in $\sum_w$. With the roll/pitch/yaw angles of the quadrotor represented by $[\phi,\theta,\psi]^T$, the rotation matrix $\boldsymbol{R}_b$ from $\sum_b$ to $\sum_w$ is given as

$$\boldsymbol{R}_b=\begin{bmatrix} c_\theta c_\psi & -c_\phi s_\psi + s_\phi s_\theta c_\psi & s_\phi s_\psi + c_\phi s_\theta c_\psi \\ c_\theta s_\psi & c_\phi c_\psi + s_\phi s_\theta s_\psi & -s_\phi c_\psi + c_\phi s_\theta s_\psi \\ -s_\theta & s_\phi c_\theta & c_\phi c_\theta \end{bmatrix}, \tag{1}$$

where c_ϕ and s_ϕ denote, respectively, $\cos\phi$ and $\sin\phi$.

In the frame system as shown in Fig. 1. The equations of position loop for the quadrotor can be written as:

$$\dot{\boldsymbol{p}}=\boldsymbol{v}, \tag{2}$$

$$m\dot{\boldsymbol{v}}=\boldsymbol{R}_b\boldsymbol{F}_{motor}+\boldsymbol{R}_b\boldsymbol{F}_{drag}+\boldsymbol{F}_{dis}+mg\boldsymbol{e}_3, \tag{3}$$

where $\boldsymbol{F}_{motor}=[0,0,-F_b]^T$ is the force applied to the vehicle expressed in $\sum_b$, $\boldsymbol{F}_{drag}=[F_{drag}^x,F_{drag}^y,F_{drag}^z]^T$ is the air drag expressed in $\sum_b$ which related to the air drag coefficient and quadrotor velocity, $\boldsymbol{F}_{dis}=[F_{dis}^x,F_{dis}^y,F_{dis}^z]^T$ is the disturbance force which may be caused by wind gusts or other environmental factors, g is the gravity constant and $\boldsymbol{e}_3=[0,0,1]^T$. The translational dynamics of the quadrotor are shown in (2)(3) and also can be represented as

$$\ddot{x}=-\frac{F_b}{m}(\sin\phi\sin\psi+\cos\phi\sin\theta\cos\psi)+\frac{F_{drag}^{xw}}{m}+\frac{F_{dis}^x}{m}, \tag{4}$$

$$\ddot{y}=-\frac{F_b}{m}(-\sin\phi\cos\psi+\cos\phi\sin\theta\sin\psi)+\frac{F_{drag}^{yw}}{m}+\frac{F_{dis}^y}{m}, \tag{5}$$

$$\ddot{z}=-\frac{F_b}{m}\cos\phi\cos\theta+\frac{F_{drag}^{zw}}{m}+\frac{F_{dis}^z}{m}+mg, \tag{6}$$

where $F_{drag}^{wx},F_{drag}^{wy},F_{drag}^{wz}$ is the air drag expressed in $\sum_w$. To decouple and simplify the dynamics (4)-(6), the feedback linearization techniques are utilized. Define virtual input as

$$u_x=-\frac{F_b}{m}(\sin\phi\sin\psi+\cos\phi\sin\theta\cos\psi), \tag{7}$$

$$u_y=-\frac{F_b}{m}(-\sin\phi\cos\psi+\cos\phi\sin\theta\sin\psi), \tag{8}$$

$$u_z=-\frac{F_b}{m}\cos\phi\cos\theta+g, \tag{9}$$

and define the lumped disturbances as

$$f_x=\frac{F_{drag}^{xw}}{m}+\frac{F_{dis}^x}{m}, \tag{10}$$

$$f_y=\frac{F_{drag}^{yw}}{m}+\frac{F_{dis}^y}{m}, \tag{11}$$

$$f_z=\frac{F_{drag}^{zw}}{m}+\frac{F_{dis}^z}{m}. \tag{12}$$

Then the dynamics (4)-(6) can be rewritten into the compact form of

$$\ddot{\boldsymbol{\xi}}=\boldsymbol{u}+\boldsymbol{f}, \tag{13}$$

where $\boldsymbol{\xi}=[x,y,z]^T$, $\boldsymbol{u}=[u_x,u_y,u_z]^T$ and $\boldsymbol{f}=[f_x,f_y,f_z]^T$. Dynamics (13) is called the simplified control design model, based on which the controller and UDE will be designed. According to [14], the following assumption on the lumped disturbances is employed.

Assumption 1: The lumped disturbances $\boldsymbol{f}=[f_x,f_y,f_z]^T$ are bounded, for $\forall t>0$.

The desired reference trajectory is given as $\boldsymbol{\xi}_r=[x_r,y_r,z_r]^T$, which is continuous differentiable and bounded up to its second-order time derivative. The overall control objectives is to design the UDE-based position controller for quadrotor system to drive the states $\boldsymbol{\xi}=[x,y,z]^T$ to track the reference signals $\boldsymbol{\xi}_r=[x_r,y_r,z_r]^T$ only with position measurements feedback.

III. Control System Design

In this section, the UDE-based position controllers will be developed by applying the passivity techniques to avoid using velocity measurements.

A. A Generic UDE-Based Control Design

We combine the classic PD controller and the UDE to dynamically compensate for the effects of the lumped

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disturbances [14]. The UDE-based controllers in the following form are considered:

$$u_i = \ddot{\xi}_{ri} + k_p\tilde{\xi}_i + k_d\dot{\tilde{\xi}}_i - \hat{f}_i,\ i \in \{x, y, z\} \tag{14}$$

where $\ddot{\xi}_{ri}$ is the feedforward term to deliver asymptotic tracking of a desired trajectory,

$$\tilde{\xi}_i = \xi_{ri} - \xi_i \tag{15}$$

are the tracking errors, k_p and k_d are the control gains, and $\hat{f}_i$ are the estimation of the UDEs. Following the idea proposed in [14], the UDEs are designed in the frequency domain to satisfy

$$\hat{F}_i(s) = G_i^f(s)F_i(s) = G_i^f(s)\left[\ddot{\Xi}_i(s) - U_i(s)\right], \tag{16}$$

where $G_i^f(s)$ are stable filters to be specified later, $\ddot{\Xi}_i(s) = \mathcal{L}\left\{\left[\ddot{\xi}_i(t)\right]\right\}$, and the Laplace transform of other variables are denoted by the corresponding capital letters. Substituting (14) into (16) yields

$$\begin{aligned}\hat{F}_i(s) &= \frac{G_i^f(s)}{1-G_i^f(s)}\left[\ddot{\Xi}_i(s) - \ddot{\Xi}_{ri}(s) - k_p\tilde{\Xi}_i(s) - k_d\dot{\tilde{\Xi}}_i(s)\right] \\ &= -\frac{G_i^f(s)}{1-G_i^f(s)}\left[k_p\tilde{\Xi}_i(s) + k_d\dot{\tilde{\Xi}}_i(s) + \ddot{\tilde{\Xi}}_i(s)\right].\end{aligned} \tag{17}$$

Select the following low-pass filter with $T_i > 0$,

$$G_i^f(s) = \frac{1}{T_i s + 1}, \tag{18}$$

and UDEs (17) become

$$\hat{F}_i(s) = -\frac{1}{T_i(s)}\left[k_p\tilde{\Xi}_i(s) + k_d\dot{\tilde{\Xi}}_i(s) + \ddot{\tilde{\Xi}}_i(s)\right]. \tag{19}$$

The time-domain expression of (19) is

$$\hat{f}_i = -\frac{1}{T_i}\left(k_d\tilde{\xi}_i + k_p\int_0^t\tilde{\xi}_i dt + \dot{\tilde{\xi}}_i\right), \tag{20}$$

which can be further substituted back to (14) leads to the UDE-based controllers

$$u_i = \ddot{\xi}_{di} + \left(k_p + \frac{k_d}{T_i}\right)\tilde{\xi}_i + \frac{k_p}{T_i}\int_0^t\tilde{\xi}_i dt + \left(k_d + \frac{1}{T_i}\right)\dot{\tilde{\xi}}_i. \tag{21}$$

where the velocity measurements are required.

B. UDE-Based Controller Applying Passivity Techniques

The passivity techniques can be adopted in the position controller to deal with the unavailabilities of $\boldsymbol{v} = \left[v_x, v_y, v_z\right]^T$. Inspired by [16], we design the controller as

$$\dot{\eta}_i = -\frac{1}{T_p}\eta_i + \frac{1}{T_p}\tilde{\xi}_i, \tag{22}$$

$$u_i = \ddot{\xi}_{ri} + k_1\tilde{\xi}_i + k_2\left(-\frac{1}{T_p}\eta_i + \frac{1}{T_p}\tilde{\xi}_i\right) - \hat{f}_i^p, \tag{23}$$

where the subscript p represents "passivity", η_i are the auxiliary variables, $\hat{f}_i^p$ are the variations of UDEs to be constructed later without using velocity measurements $\boldsymbol{v} = \left[v_x, v_y, v_z\right]^T$, and $T_p > 0$. Compared with (21), controller design (23) avoids the use of velocity feedback terms and both the nominal control and the UDE. In fact, we have the following relationship in the frequency domain,

$$\mathcal{L}^{-1}\left\{\left[-\frac{1}{T_p}\eta_i + \frac{1}{T_p}\tilde{\xi}_i\right]\right\} = \frac{s}{T_p s + 1}\tilde{\Xi}_i(s) \tag{24}$$

where $\dfrac{s}{T_{pi}s+1}$ is the passive filter in the frequency domain. In this design, the term $k_2\left(-\dfrac{1}{T_p}\eta_i + \dfrac{1}{T_p}\tilde{\xi}_i\right)$ in controller (23) acts as a damping, which replaces the unavailable velocity feedback $k_2\dot{\tilde{\xi}}_i$ and do not force the velocity motions to settle down at the origin.

Similarly to (16), the UDEs are designed as

$$\hat{F}_i^p(s) = G_i^{fp}(s)F_i(s) = G_i^{fp}(s)\left[\ddot{\Xi}_i(s) - U_i^p(s)\right], \tag{25}$$

Which when substituted with (23) become

$$\hat{F}_i^p(s) = \frac{G_i^{fp}(s)}{1-G_i^{fp}(s)}\left[\ddot{\Xi}_i(s) - \ddot{\Xi}_{ri}(s) - k_1\tilde{\Xi}_i(s) - \frac{k_2 s}{T_p s + 1}\tilde{\Xi}_i(s)\right]. \tag{26}$$

Since $\ddot{\Xi}_i(s)$ are not measurable, we utilize the following relationships

$$\begin{aligned}\ddot{\Xi}_i(s) &= s\dot{\Xi}_i(s) - \dot{\xi}_i(0) \\ &= s^2\Xi_i(s) - s\xi_i(0) - \dot{\xi}_i(0).\end{aligned} \tag{27}$$

Note that $G_i^{fp}(s)$ are required to make (26) physically realizable. Consider the relationships (25)(26), which require that $\dfrac{G_i^{fp}(s)}{1-G_i^{fp}(s)}$ are stable and have relative orders no less than 2. The lowest-order design of $G_i^{fp}(s)$ that satisfies the aforementioned requirements is

$$G_i^{fp}(s) = \frac{1}{T_{i1}s^2 + T_{i2}s + 1}, \tag{28}$$

where $T_{i1} > 0$ and $T_{i2} > 0$. The corresponding UDE design is given by

$$\hat{F}_i^p(s) = \frac{1}{T_{i1}s + T_{i2}}\left[s\Xi_i(s) - \xi_i(0) - \frac{\dot{\xi}_i(0)}{s} - \dot{\Xi}_{ri}(s) - \left(\frac{k_1}{s} + \frac{k_2}{T_p s + 1}\right)\tilde{\Xi}_i(s)\right], \tag{29}$$

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Which can be further substituted back to (23) leads to the proposed UDE-based controllers.

C. Proposed Cotroller in Time-Domain

In the proposed method, part of the control law (24)(29) are designed in frequency domain. To program these estimators in C language, we need to translate them into a discrete-time form. Define a high-pass filter $G_i^{HPF1}(s)=\frac{s}{T_{i1}s+T_{i2}}$ and low-pass filters $G_i^{LPF1}(s)=\frac{1}{T_{i1}s+T_{i2}}$ and $G_i^{LPF2}(s)=\frac{1}{T_p s+1}$, and these three filters in the discrete-time domain can be expressed as

$$\mathrm{HPF1}(u_k)=y_k=\frac{1}{T_{i1}+T_{i2}\delta_t}(T_{i1}y_{k-1}+u_k-u_{k-1}), \tag{30}$$

$$\mathrm{LPF1}(u_k)=y_k=\frac{1}{T_{i1}+T_{i2}\delta_t}(T_{i1}y_{k-1}+\delta_t u_k), \tag{31}$$

$$\mathrm{LPF2}(u_k)=y_k=\frac{1}{T_p+\delta_t}(T_p y_{k-1}+\delta_t u_k), \tag{32}$$

where u_k is the input at k step, y_k is the output at k step, δ_t is the time interval. Then the estimator (29) in the time domain is expressed as

$$\hat{f}_i^p(t)=\mathrm{HPF1}(\xi_i)-\mathrm{LPF1}\left(\dot{\xi}_{ri}+k_1\int\tilde{\xi}_i dt+k_2 LPF2(\tilde{\xi}_i)\right). \tag{33}$$

Define a high-pass filter $G_i^{HPF2}(s)=\frac{s}{T_p s+1}$, the passive filter (24) in time-domain is expressed as

$$\dot{\eta}_i(t)=\mathrm{HPF2}(\tilde{\xi}_i). \tag{34}$$

With (22)(33)(34), the controller (23) in time domain can be expressed as

$$\begin{aligned}u_i=&\ddot{\xi}_{ri}+k_1\tilde{\xi}_i+k_2\mathrm{HPF2}(\tilde{\xi}_i)-\mathrm{HPF1}(\xi_i)\\&+\mathrm{LPF1}\left(\dot{\xi}_{ri}+k_1\int\tilde{\xi}_i dt+k_2\mathrm{LPF2}(\tilde{\xi}_i)\right).\end{aligned} \tag{35}$$

IV. Stability Analysis

The closed-loop system consisting of (13)(22)(23) can be rewritten into a compact from as

$$\begin{bmatrix}\dot{\tilde{\xi}}\\ \ddot{\tilde{\xi}}\\ \dot{\eta}\end{bmatrix}=\underbrace{\begin{bmatrix}0 & 1 & 0\\ -k_1-\frac{k_2}{T_p} & 0 & \frac{k_2}{T_p}\\ \frac{1}{T_p} & 0 & -\frac{1}{T_p}\end{bmatrix}}_{A}\begin{bmatrix}\tilde{\xi}\\ \dot{\tilde{\xi}}\\ \eta\end{bmatrix}+\underbrace{\begin{bmatrix}0\\1\\0\end{bmatrix}}_{B}\tilde{f}_p, \tag{36}$$

where $\tilde{f}_p=\hat{f}_p-f_p$. To analysis the stability of system (36), we compute the characteristic polynomial of $\boldsymbol{A}$ as

$$|s\boldsymbol{I}-\boldsymbol{A}|=s^3+\frac{1}{T_p}s^2+\left(k_1+\frac{k_2}{T_p}\right)s+\frac{k_1}{T_p}. \tag{37}$$

By applying Routh criterion, we readily derive the following stability condition

$$\begin{cases}k_1>0\\ k_2>0\end{cases}. \tag{38}$$

Under this condition (38), system (36) with $\tilde{f}_p$ as input is globally input-to-state stable and input bounded-state stable.

Remark 1: The major advantages of using passivity techniques are two folds. First, the ability of responding to a time-varying command is still retained in (23), compared with the case where the controller (14) is used with $\dot{\xi}_{ri}=0$. Second, unlike the differentiator whose performance relies on a small parameter T_{dif}, the passive filter parameter T_p does not need to be small to deliver a good tracking performance [16]. Therefore the passivity-based controller is less sensitive to the noises in $\boldsymbol{\xi}=[x,y,z]^T$ than controller (14) with the differentiators.

V. Simulation and Experimental Results

In this section, the proposed approach is validated in Gazebo simulation and flight experiment.

A. Simulation and Experimental Setup

The proposed approach in discrete time (35) is programmed in C language based on Prometheus opensource project [17]. The simulation is conducted on PX4 software-in-loop (SITL) [18] with Gazebo simulation platform and the experiment is conducted on the quadrotor as shown in Fig. 2. The position measurements are obtained from GPS in simulation and Vicon system in experiment. The onboard computer (Nvidia TX2) runs the proposed controller and Mavros package to send the control command to the Pixhawk Autopilot. The codes are published in Github and can be founded at https://github.com/amov-lab/Prometheus.

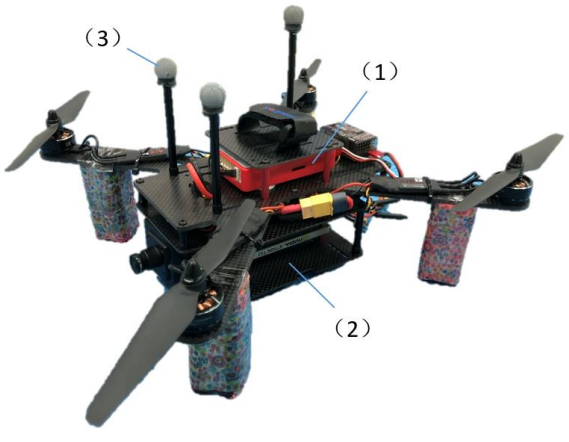


Fig. 2. The quadrotor platform : (1) Pihawk Autopilot (2) Nvidia TX2 (3) Vicon markers.

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The controller feedback gains are chosen as

$$\begin{cases} k_1 = 0.5, \ \ k_2 = 2.0 \\ \quad T_p = 0.2 \\ T_1 = 2.0, \ \ T_2 = 1.0 \end{cases}.$$

B. Simulation Results

TABLE I. CASE STUDIES IN THE SIMULATIONS

Cases	Scenarios	Disturbance
1	Hovering	×
2	Hovering	√
3	Trajectory tracking	×

Based on the current common verification scenarios, simulation calculations on hover, anti-jamming and fly-around were carried out to verify the algorithm mentioned above.The case studies considered in the simulations are summarized in Table I. The simulation results are shown form Fig. 3 to Fig. 6.

1) Case 1: The simulation aims to make the target hover on the fixed point.The time length of simulation is about 60 seconds.The horizontal position refernces x_r and y_r are set to 0 m; the altitude reference z_r is set to -1.0 m. Fig. 3 shows the position trajectories in Case 1 (the black line is the reference signal, the blue line is the measurements).

2) Case 2: To test and verify the anti-jamming capability of the proposed controller, a random input disturbances is added to the quadrotor x/y motion at $t = 10s$ and disabled at $t = 35s$. Fig. 4 shows the position trajectories in Case 2. The target can maintain stable after disturbance applied and recover to the desired point within 6-7 s. Fig. 5 shows the forces generated by motors.

3) Case 3: The control obejective of this simulation is to make the quadrotor track a circular trajectory. The radius of the circle is set to 2 m and the reference linear velocity is set to 0.5 m/s. Fig. 6 shows the position trajectories in Case 3.

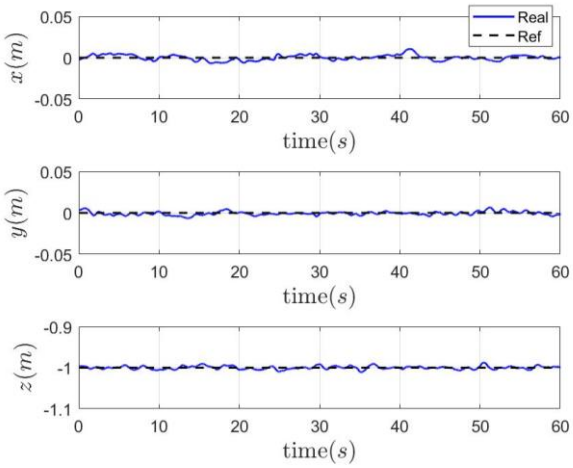


Fig. 3. Case 1: The position trjectories (Hovering).

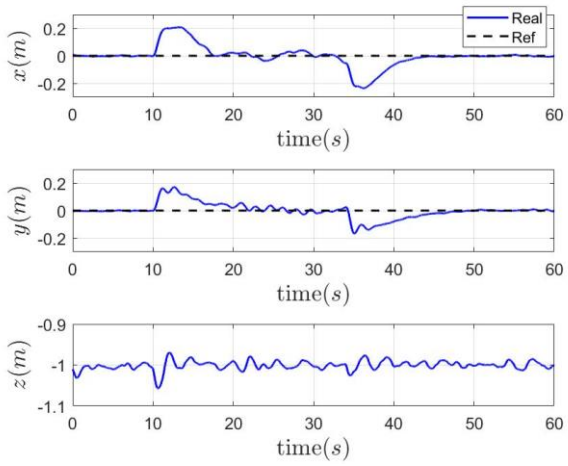


Fig. 4. Case 2: The position trjectories (Hovering with disturbances).

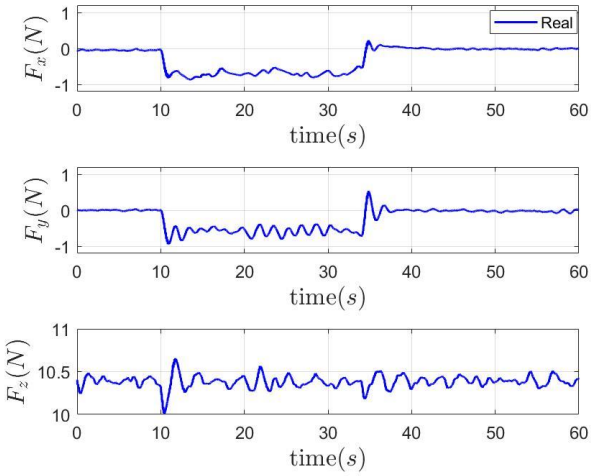


Fig. 5. Case 2: The forces generated by motors (Hovering with disturbances).

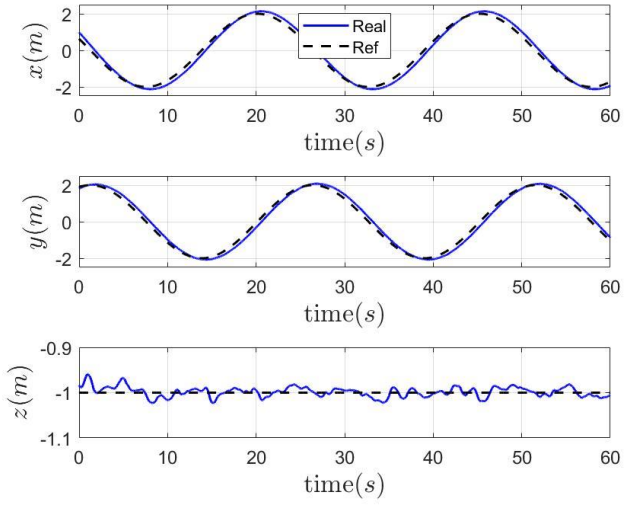


Fig. 6. Case 3: The position trjectories (Trajectory tracking).

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C. Experimental Results

1) Case 4 (Hovering): The experiment setup is same as the Case 1. Fig. 7 shows the position tracking error.

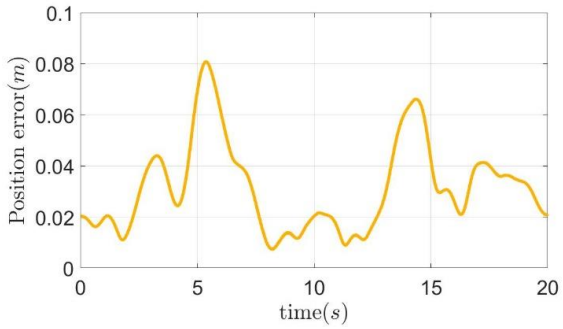


Fig. 7. Case 4: The position error (Hovering).

2) Case 5 (Trajecotry tracking): The experiment setup is same as the Case 3. Fig. 8 shows the X-Y plane trajectory.

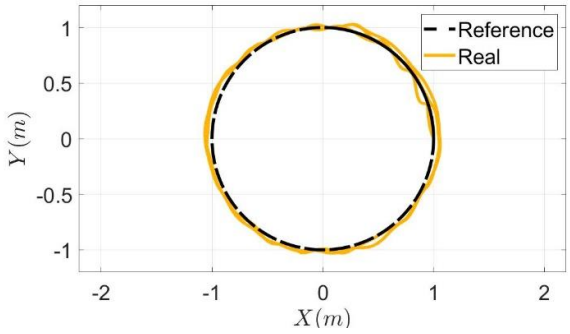


Fig. 8. Case 5: The X-Y plane trajecotry (Trajectory tracking).

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